

Time Frequency Analysis of Non-Stationary Motor Imagery EEG Signals

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Abstract

Brain Computer Interface (BCI) offers an effective medium of communication for physically disabled people who are partially or completely locked in their body. It also improves the performance level of healthy people by providing them a new input modality. Recently, BCI systems based on motor imagery electroencephalogram (EEG) signals have been focussed intensively. Extraction and classification of motor imagery features is the most crucial part of motor imagery based BCI system. The biggest challenge in processing motor imagery EEG signals is that these signals are non stationary. In the present investigation, left hand and foot motor imagery tasks have been classified. Time-frequency features have been extracted using discrete wavelet transform (DWT) to deal with the non stationarity of motor imagery EEG signals. Linear discriminant analysis (LDA) is used to reduce the dimensionality of data. The efficacy of DWT and LDA has been tested using state vector machine (SVM), k- nearest neighbour (k-NN) and artificial neural network (ANN) classifiers. Overall, combining DWT with LDA and ANN achieved a promising average accuracy of 90% in 4.453 secs. The present study may prove to be helpful in developing intelligent devices to help people in locked-in state for controlling some application by imagination of the movement of their body parts.

Keywords: *Brain computer interface, EEG, Non-stationarity, DWT, SVM, k-NN, ANN.*

1. Introduction

Brain Computer Interface (BCI) has introduced a new modality in the usage of computer. BCI uses signals from the human brain and controls a computer system. It establishes a direct communication channel between brain and computer system. A BCI system can be used to help physically disabled people by translating their thoughts into commands. This technology has shown outstanding results even for able bodied people. It can be viewed as a sort of mind reading where computer detects user's intention and works accordingly. The BCI pipeline consists of a number of stages which

includes data recording, feature extraction, dimensionality reduction, training, classification and control application as shown in Figure 1 [1-4]. Recently, BCI systems based on motor imagery EEG signals have been focussed intensively [5-8]. Electrical activity of brain is a real time representation of the brain. Imagined movements result in brain activities which are to be recognised by the classifier. Motor Imagery is a higher level feature which is currently being used in BCI. The planning and execution of body movements are directly linked to pre-movement attenuation and post-movement rise in amplitude of brain rhythms. Event related desynchronisation (ERD) and event related synchronisation (ERS) are two brain signals which are generated in response of motor imagination. ERD is suppression in synchronisation of mu signals which occur during movement imagination when movement is planned. ERS is increase in synchronisation of mu signals which occur at the end of imagined movement. In combination, these signals provide the information about the body part whose movement is imagined [9].

Motor Imagery EEG signals are statistically non-stationary. The signals have specific information in both the temporal and frequency domain. Moreover, the amplitude and frequency of these signals change from one state to another and from one user to other. The characteristics of these values also change with age [9]. Time domain and frequency domain representations do not show the characteristics of EEG signals accurately even when integrated over the observation interval. A combined approach *i.e.* time-frequency domain analysis provides a solution to this problem. The time-frequency domain representation captures sudden temporal variations of the signals while still keeping frequency domain information [10].

In the present investigation, non-stationary features associated with left hand and foot motor imagery tasks have been extracted and classified. Time-frequency motor imagery features have been extracted using DWT to deal with the non stationarity of motor imagery EEG signals. LDA is used to reduce the dimensionality of data. The efficacy of DWT and LDA has been tested using SVM, k-NN and ANN classifiers.

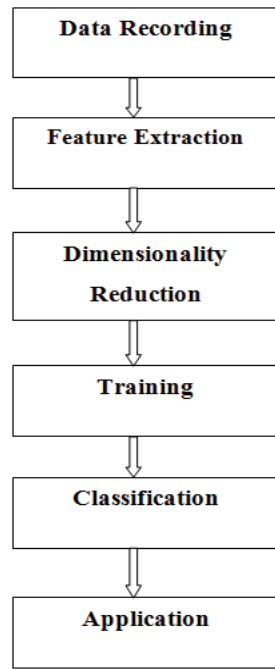


Figure1: Framework of a BCI System

After brief overview of BCI system, the motivation to carry out present investigation has been described in the succeeding section.

1.2 Motivation of Present Investigation

Neuromuscular channels of human body are directed and controlled by the brain. Many disorders disrupt the normal activities of the neuromuscular channels. This leads to impairment of body muscles and disable the person from living a healthy life. BCI systems provide a new channel of communication to the patients with motor disabilities. BCI users are categorised into three classes. First class consists of people who still retain control of speech and/or hand. Second class includes people who retain very little neuromuscular control such as eye movement. Third class consists of people who become completely locked in their body. People of first and second category may use speech or eye movement to control and operate some devices. But the people of third category require deciphering of thoughts. MI based BCI serve the third class of people but is limited in its current accuracy and speed. The present investigation is an effort to improve the accuracy and speed of MI based BCI system [11].

2. Related Work

In this section, the relevant studies conducted by various authors in context of this work have been discussed. Moreover, research gap identified by the authors are presented that motivates to carry out the

present study. The relevant studies in this context are as follows:

1. Stefan Cocoschi *et al.* (2006) presented the use of self organising fuzzy neural network to classify time domain features of movement imagination signals and suggested further improvement in classification accuracy [12].
2. Plamen Doynov *et al.* (2008) classified MI tasks using fisher discriminant analysis (FDA) using wavelet based feature extraction method. Concatenation of features is suggested to improve the performance of the BCI system [13].
3. Fabien Lotte *et al.* (2009) extracted MI EEG features using CSP, filter band CSP and band power and classified the obtained features using LDA. Authors suggested the use of fuzzy inference systems to improve the performance [14].
4. Hongyu Sun *et al.* (2010) classified motor imagery tasks by SVM using common spatial pattern (CSP) and independent component analysis (ICA) as feature extraction methods. The importance of relevant features in achieving accurate classification has also been discussed [15].
5. Rajesh P. N. Rao *et al.* (2010) used ERS and ERD features to classify non stationary EEG signals by LDA classifier. Optimisation of signal processing techniques has been advised by the authors [16].
6. Saugat Bhattacharyya *et al.* (2010) implemented LDA, quadrature discriminant analysis (QDA) and k-NN on wavelet coefficients to classify imagery tasks. Authors stressed the use of time frequency methods of analysis to deal with the non stationarity of EEG signals [17].
7. Yu Zhou *et al.* (2013) demonstrated the use of short time fourier transform (STFT) as feature extraction method and classified left and right hand movement imagination by back propagation neural network. Signal processing methods in time frequency domain have been considered advantageous [18].
8. Roxana Aldea *et al.* (2014) explored the efficiency of wavelet coefficients as input features to LDA classifier to classify left or right hand movement imagination and rest state. The multi dimensional feature extraction has been highlighted [19].
9. A. Sivakami *et al.* (2015) classified left and right hand movement imagination using back propagation neural network. ERD quantification, FFT power spectrum and DWT have been used as feature extraction methods. The authors suggest time frequency analysis methods to be suitable for motor imagery signals [20].

After an outlook of the literature presented here and keeping in view the research gap identified certain objectives have been set for the present study as discussed in forthcoming section.

3. Objectives of the Study

The key research objective that has been focussed in this investigation is to analyse non-stationary motor imagery EEG signals using time-frequency analysis techniques. The objectives of the present study are as follows:

- To transform raw motor imagery EEG signals using time frequency analysis method.
- To extract relevant features from transformed motor imagery EEG signals.
- To classify of motor imagery EEG signals using various classifiers by utilising extracted features.
- To compare classifiers on the basis of classification accuracy and classification time.

To achieve the overall objective the methodology used in the present investigation are presented in the proceeding section.

4. Methods and Materials

This section describes the dataset, features, feature extraction method and classifiers that have been used in the present study.

4.1 Dataset

Publically available online dataset, “Dataset 1 (Motor Imagery) of BCI competition IV” has been used for the investigation. BrainAmp MR plus amplifiers and an Ag/AgCl electrode cap had been used for recording the dataset. Signals from 59 EEG positions that were most densely distributed over sensorimotor areas have been measured and recorded. Signals have been band-pass filtered between 0.05 and 200 Hz frequency band and then digitized. In this study, two classes of motor imagery *i.e.* left hand and foot have been considered for analysis and classification [21]. Figure 2 displays a sample of dataset which has been used in the present investigation. The plots of raw left hand and foot motor imagery datasets have been shown in figure 3 and 4 respectively.

	1	2	3	4	5	6	7	8	9	10
1	AF3	AF4	F3	F3	F1	F1	F2	F4	F6	FC3
2	-3	-3	-3	-8	-4	-5	-4	1	1	-6
3	-25	-21	-26	-63	-26	-26	-28	12	10	-43
4	-34	-15	-52	-97	-22	-44	-22	60	49	-85
5	24	49	-30	6	59	48	60	129	88	-59
6	44	37	-15	54	95	80	73	124	47	-48
7	30	-16	-5	28	88	56	45	100	23	-128
8	74	39	60	87	129	114	112	166	107	-53
9	105	93	107	116	157	150	162	225	163	197
10	105	66	106	124	157	134	137	196	120	139
11	89	16	108	119	118	83	78	135	76	54
12	52	-8	90	66	60	25	31	104	69	43

Figure 2: A sample of dataset

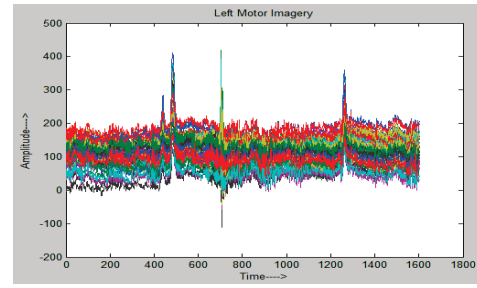


Figure 3: Plot of Raw Left Hand Motor Imagery Dataset

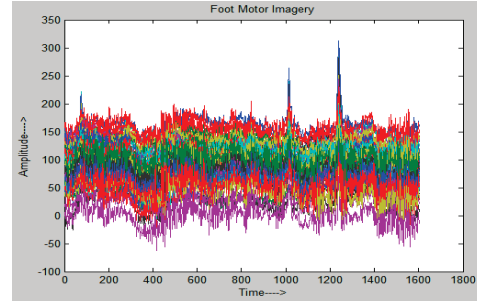


Figure 4: Plot of Raw Foot Motor Imagery Dataset

4.2 Analysis using DWT

Choice of feature extraction algorithm has more influence on final performance of BCI system as compared to classification algorithm. So, it is recommended to extract good features to make the task of classifier easy. If the features are irrelevant and do not describe the mental state of the user well, the classification algorithm will not be able to produce the accurate results. Consequently, it would lead to low recognition rate at the output of classifier which will make it difficult for the user to use the system.

Motor imagery EEG signals possess non-stationary and transitory characteristics. Pure frequency domain methods of analysis lead to losing some important information hidden in these signals. Hence, time-frequency methods have to be used to obtain accurate results. The time frequency methods transform the raw EEG data into a view where amplitude is represented at a certain time and frequency. This makes it easy to identify the amount of brain activity at certain frequency interval and at a certain time. In this study, Wavelet transform has been used as it has been emphasized that it is suitable for analysis of non-stationary signals and prove to be advantageous over pure frequency domain methods [22-24]. Wavelet transform decomposes a signal into a set of wavelets $\phi_{a,b}$. Each wavelet $\phi_{a,b}$ is scaled and translated version of same wavelet ϕ known as mother wavelet which is defined using eq. (1).

$$\phi_{a,b} = \frac{1}{\sqrt{a}} \phi\left(\frac{t-b}{a}\right) \quad (1)$$

The wavelet transform of a signal $x(t)$ is defined using eq. (2).

$$w_x(a,b) = \int_{-\infty}^{+\infty} x(t) \phi_{a,b}(t) dt \quad (2)$$

Using wavelets, high frequencies can be analysed with high temporal resolution and low frequencies can be analysed with a high frequential resolution. In order to process digital signals, a discrete approximation of wavelet coefficient is obtained. DWT based analysis is based on filter banks. Two digital filters and two down samplers at every stage decompose the signal into its spectral components [25]. Here, motor imagery EEG signals are decomposed into frequency sub bands using DWT. A set of statistical features which represent the wavelet coefficients have been extracted from the sub bands. The number of level of decomposition is based on dominant frequency of the signal. EEG signals have useful frequency components below 30 Hz. So, the number of levels of decomposition is chosen to be 5. The plots of wavelet transformed datasets have been shown in figure 5 and 6 respectively.

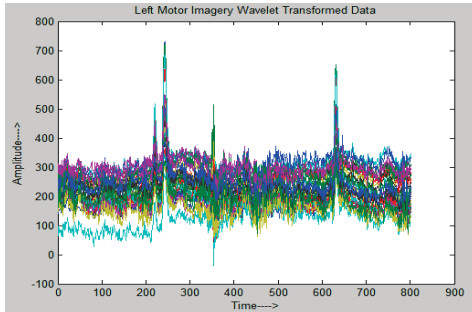


Figure 5: Plot of Wavelet Transformed Left Motor Imagery Dataset

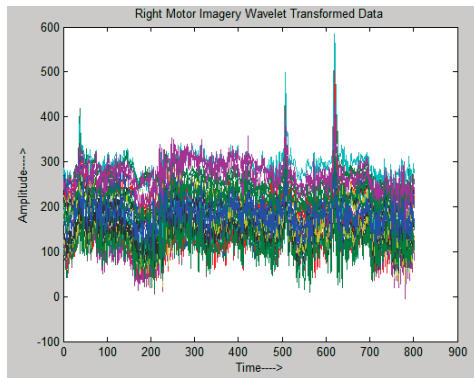


Figure 6: Plot of Wavelet Transformed Foot Motor Imagery Dataset

4.3 Feature Extraction

The feature vector obtained from motor imagery EEG signals is of very high dimensionality. This problem is generally known as curse of

dimensionality. To tackle this problem, dimensionality reduction methods have to be used [26, 27]. These methods make it possible to work with a set of features which are small in size and describe some relevant properties of signal which often leads to better performance of overall BCI system [28, 29]. In this work, LDA has been used as dimensionality reduction method. Figure 7 and 8 respectively show the LDA transformed datasets. It is a supervised method of transformation which uses class separability as criteria, maximises the distance between class means and projects d-dimensional space onto a new k- dimension space where $k < d$. It uses a linear hyperplane to separate data between two classes. The class assigned to the data depends on side of hyperplane it falls in.

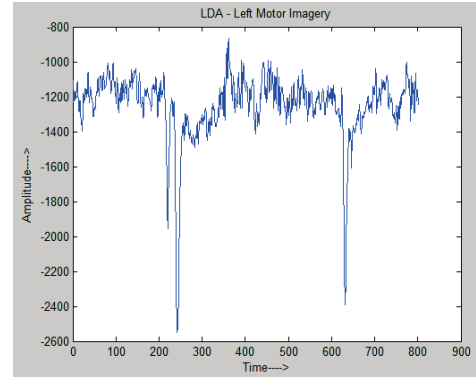


Figure 7: Plot of LDA Transformed Left Hand Motor Imagery Dataset

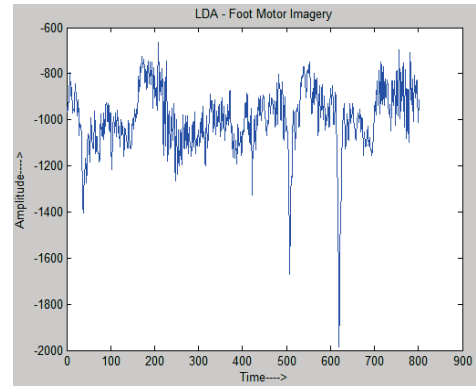


Figure 8: Plot of LDA Transformed Foot Motor Imagery Dataset

4.4 Training and Classification

The task of a classifier is to assign each feature to a class using the feature vector provided by the feature extractor [30, 31]. In this study, two class classification (left hand and foot motor imagination) has been investigated. An SVM classifier with RBF kernel, k-NN classifier with euclidean distance metric and multilayer feed forward neural network trained by back

propagation algorithm optimised using levenberg marquardt optimisation technique (as shown in figure 9) have been used to classify time frequency motor imagery EEG features of motor imagination tasks extracted using DWT.

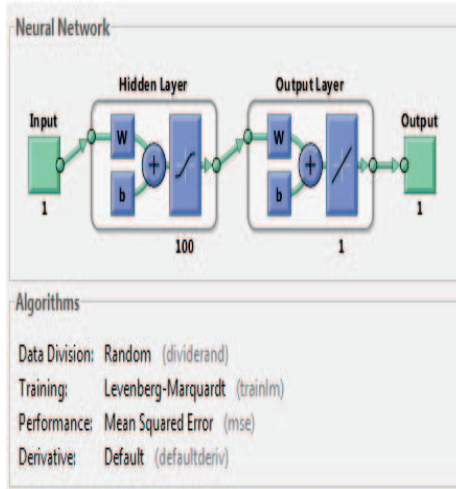


Figure 9: Neural Network Trained for Classification

5. Implementation of the present Investigation

In the present study, EEG signals of left hand and foot motor imagination of 8 subjects have been used to test the efficacy of DWT by using SVM, k-NN, ANN classifiers. EEG signals have been decomposed using discrete wavelet transform and feature set has been extracted from sub bands. The dimension of thus extracted features has been reduced using LDA. Classifiers based on SVM, k-NN and ANN have been trained using these input features. For developing classifiers, 800 samples have been taken from 1600 samples for training and remaining 800 examples have been used for testing the developed classifiers as shown in Table 1. The same procedure has been repeated for the dataset from 8 subjects.

Table 1: Class distribution of samples in the training and test set

Class	Training Set	Test Set	Total
Left Hand Motor Imagination	800	800	1600
Foot Motor Imagination	800	800	1600
Total	1600	1600	3200

To measure the performance of the classifiers mean square error (MSE) has been calculated using eq. (3).

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2 \quad (3)$$

Where, $\hat{Y}_i$ = Vector of n predictions made by classifier

Y_i = Observed values to the input to the classifier which predicts class of the input

The procedure of the present investigation has been summarised as follows:

- The features have been calculated in time frequency domain using DWT.
- The feature size has been reduced using LDA algorithm which removes the irrelevant features which are redundant.
- The classification of motor imagery tasks is carried out using SVM, k-NN and ANN classifiers.

The results obtained by the implementation of above procedure have been presented in the proceeding section.

6. Results and Discussions

The results obtained by the present investigation have been shown in Table 2. Results have been recorded for all the subjects individually and then averaged to obtain the final result. As evident from Table 2, the average classification rate obtained using SVM classifier is 86.88%, k-NN classifier is 78.75% and ANN classifier is 90%. Moreover, average training and classification time taken by SVM is 0.969 seconds, k-NN obtains its result in 0.605 seconds and ANN requires 4.453 seconds.

Table 2: Statistical parameters of SVM, k-NN and ANN classifiers used for classification

Feature Extraction Method	Classification Method	Average Accuracy (%)	Average Training and Classification Time (Seconds)
DWT + LDA	SVM	86.88	0.969
	k-NN	78.75	0.605
	ANN	90	4.453

In literature, Stefan Cosochi [12] used self organising fuzzy neural network to classify time domain features of movement imagination signals with an accuracy of 82.68% in 3.62 seconds. Plamen Doynov [13] classified MI tasks using fisher discriminant analysis (FDA) using wavelet based feature extraction method and achieved 75% accuracy. Fabien Lotte [14] classified MI EEG signals by LDA using CSP, filter band CSP and band power as features and achieved an accuracy of 60-65%. Hongyu Sun [15] classified motor

imagery tasks by SVM using common spatial pattern (CSP) and independent component analysis (ICA) as feature extraction methods and achieved an accuracy of 90% using CSP and 84% using ICA. Rajesh P. N. Rao [16] classified non stationary EEG signals using ERS and ERD features by LDA and achieved an accuracy of 80%. Saugat Bhattachayya [17] used LDA, quadrature discriminant analysis (QDA) and k-NN on wavelet coefficients and achieved an accuracy of 80%. A. Sivakami [20] used back propagation neural network to classify left and right hand movement imagination and achieved an accuracy of 70% for right imagination and 87% for left imagination. Although, previous studies in literature have shown good performance on classification of MI EEG signals, the present investigation has achieved higher classification accuracy and smaller classification time. The present investigation combines DWT and LDA to produce relevant and significant features. The high accuracy achieved using ANN demonstrates that the features obtained using DWT combined with LDA well represent the non-stationary EEG signals. By the usage of these features, a good level of discrimination between MI tasks can be achieved.

7. Conclusion

Although, valuable and promising achievements have already been obtained in the field of BCI, it is still in its infancy. A lot of work has yet to be done to make BCI a technology of public use. A BCI system which makes fewer mistakes would be more convenient for users. A system which commits fewer mistakes takes lesser time to correct them. So, such system would also have a higher classification speed. Hence, with accuracy, speed of the system also increases. The major challenge which limits the classification accuracy and hence, the speed of motor imagery based BCI system is that motor imagery EEG signals are non stationary in nature.

In the present study, non stationarity of motor imagery EEG features has been dealt by using time frequency analysis. Motor imagery EEG signals have been decomposed into time frequency representation using DWT and statistical features have been extracted from DWT sub bands of EEG signals. Dimensionality of the features extracted using DWT has been reduced using LDA. Three classifiers viz. SVM, k-NN and ANN have been trained and compared in terms of their classification accuracy and training time relative to the observed motor imagination patterns. The results obtained by the present investigation reveal that ANN classifier is the most appropriate for motor imagery based BCI system. In summary, combining DWT with LDA and ANN achieved a promising average accuracy of 90% in 4.453 seconds. The result obtained is highly encouraging

to create a BCI system. It can be emphasized that DWT coefficients when used as features represent the motor imagery EEG signals well. These features provide a good discrimination between classes. In nutshell, the present investigation plays pivotal role to develop intelligent devices which may help physically disabled people to control some application by imagination of the movement of their body parts.

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